

VARPA: A Variable Power Amplifier

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Abstract - A 0 to 1 watt (30 dBm) software adjustable class-AB radio frequency (RF) power amplifier (PA) operating in the very high frequency (VHF) band is presented with closed-loop power control. The system uses a directional coupler and an RMS (Root-Mean Squared) logarithmic power detector, in a feedback configuration, to adjust the RF output power by step sizes of approximately 0.5 dB.

Keywords - rf power amplifier, class-ab, variable power, logarithmic power detector, directional coupler

I. INTRODUCTION

Having the capability to vary the RF output power of a wireless transmitter has become a key requirement for users and, by extension, RF system designers. This requirement is primarily driven by external factors such as energy efficiency or RF output power constraints. Furthermore, wireless systems operating in the VHF band are of general interest for long-range, license-free telemetry, since the VHF band offers a unique set of advantages over more traditional, high-frequency bands such as UHF. These advantages include: design using discrete/lumped components [1] and verification using low-cost RF test equipment.

The RF power amplifier design for VARPA emphasizes efficiency and linearity while balancing the trade-off between power step size and system cost. To this end, a closed-loop control system is designed and implemented that uses a directional coupler and an RMS logarithmic power detector, in a feedback configuration, to allow the RF output power to be adjusted in step sizes of approximately 0.5 dB. Two power control strategies were investigated, namely: variable inline attenuation, which involves placing a variable digital attenuator at the input of the PA, as well as drain bias control which involves varying the supply voltage to the RF power amplifier itself.

Due to the nature of VHF electromagnetic propagation, the system, when integrated with a radio transceiver, is ideal for use in large open spaces, such as those found in rural areas. The benefits include being able to maximize the power efficiency of the system, as the cost of utility power is generally a limiting factor for rural communities. The system could also potentially be used in community base stations for metering applications or to transmit sensor data from weather beacons, for example.

II. LITERATURE REVIEW

Using a well-characterized and proven PA design technique is a critical requirement for a robust PA design. Historically, two main methods of PA design were in widespread use. In the first method, a passive variable impedance network or mechanical 'tuner' (typically a coaxial stub tuner) is placed at the input and output ports of the RF power transistor or DUT. The impedance of these tuning networks is then manually varied until a specific operating condition (i.e. output power or efficiency) is obtained through direct measurement of the DUT in a process known as 'load-pull'.

With more research on PA design being conducted over time, newer methods have emerged that take advantage of the exponential increase in computing power over the last decade (Moore's law). Under large signal conditions, the performance (i.e. gain, efficiency) of a power amplifier depends not only on the frequency and bias set point of the power transistor, but also on the input/output power and PA loading conditions [2]. The load-pull technique has emerged as the most widely used design technique to optimize PA performance in the non-linear domain. Recent advances in commercial automatic load-pull sets and RF simulation tools have significantly simplified the traditional RF power amplifier design process. In particular, the emergence of the harmonic balance (HB) method and its mathematical variants has allowed for the development of large-signal simulated load-pull techniques [3], which have largely replaced traditional mechanical or automatic load-pull test equipment when an accurate RF power transistor software model is available to a designer. In instances where a particular RF power transistor software model is not available, a physical load-pull set is often required for a PA design.

A low cost, mechanical impedance tuner utilizing a 3 dB quadrature hybrid directional coupler and two shorted microstrip transmission line segments has been proposed in the literature [4]. The design is simple to implement, but the physical length of the transmission lines and coupler is dependent on the design frequency of the impedance tuner which makes the tuner more suitable for high band VHF and UHF load-pull.

Traditionally RF power control has been achieved by varying the supply voltage or input attenuation to a RF power amplifier [5]. Gate bias modulation is another option but introduces con-

trol stability issues due to the movement of the RF transistor's Q-point. This effects gain and stability making power control more complicated using this technique. Both traditional methods of power control offer different tradeoffs in terms of PAE and gain.

III. METHOD

The prototype system consists of four distinct modules. These include the power supply module, which provides the required power to the rest of the system and regulates the PA supply voltage. The transceiver module, which generates the required PA input drive. The power amplifier module which regulates the RF output power of the system, as well as an auxiliary controller module which acts as the global system coordinator.

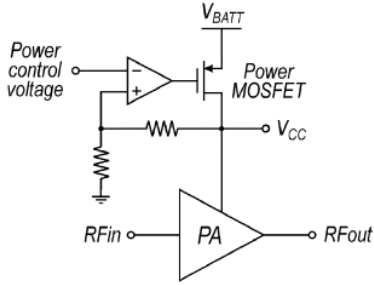


Figure 1. Proposed PA supply regulator circuit [6].

Figure 1 depicts the drain regulation circuit which is utilized in VARPA. This is a linear regulator which reduces AM sideband distortion in the RF output signal due to noise coupling into the supply rail. Although the use of the linear regulator improves signal integrity, the dissipative losses associated with the regulator can become significant at which point a switching regulator is more suitable.

The transceiver module makes use of the +20 dBm Si4468 FSK transceiver IC by Silicon Labs in order to provide the correct input drive to the PA when the IC is operated in CW mode. The IC features a class-E PA output path which is matched to it's optimum load impedance as well as a differential LNA which requires an integrated matching and balun circuit to achieve the optimum noise figure (NF) as well perform the single-ended to differential signal conversion from the antenna.

The RF PA module is built around the AFT05MS004 LDMOS RF MOSFET by NXP which requires a 7.5 V source to generate 36 dBm (4W) of RF output power. Although Gallium Nitride (GaN) transistors have largely replaced LDMOS, LDMOS is still used for VHF applications due to it's low cost. Through a simulated load pull and stability analysis of the LDMOS transistor it was found that an optimum load impedance of (7.78 – 11.26j) Ω was required in order to achieve the rated output power. Additionally harmonic suppression of 50 dBc was required in order to meet emission requirements (FCC, EU CEPT). To achieve both requirements a 2 element LC matching network and fifth order LC low-pass π -filter was designed for the PA output path.

For the system controller, a 16 MHz ATmega328 8-bit microcontroller was utilized, together with 12-bit DACs and ADCs which were used to monitor and set RF PA bias voltages.

Finally, a drain current monitor circuit for the PA, required to set the quiescent (Q) point of the PA, consisting of a differential amplifier (op-amp) and current shunt was designed and integrated into the prototype system.

IV. DESIGN CONSIDERATIONS

A number of important design decisions were taken early on to ensure that the project remained low cost and easy to design and model. For example, load-pull simulations are traditionally performed using active or passive load pull instruments that are extremely costly. Therefore, emphasis was placed on choosing a low cost RF transistor with a publicly available non-linear simulation model. This allowed the design to be refined through simulation rather than through empirical or experimental methods.

Additionally, the use of 12-bit ADC/DAC ICs was chosen as a tradeoff between step resolution and component cost. This allowed the system to be both responsive and accurate enough to satisfy user requirements. The power control algorithm used was a basic step ramp that would linearly increase the output power until a certain threshold level in dBm was met. This method provided power settling times less than 1 second, which was another requirement for the system. Another important design decision for VAPRA was to design around the 169.4 MHz license-free VHF band which is available in Europe and South Africa which allows for up to 500 mW of ERP RF output.

Low pass LC π -filter networks of order $n=5$ were also designed and integrated into all PA output paths to ensure sufficient harmonic attenuation of around 50 dBc, which is a general guidelines derived from US FCC regulations. Finally, 57 dB RMS power detection ICs were chosen as the main power sensing elements, which provided sufficient dynamic range and a linear input to output voltage transfer function for accurate power detection across the 0-30 dBm output range.

V. RESULTS

Figure 2 depicts the output vs input power curve which results when a variable attenuator is placed at the input of the PA, while Figure 3 is a plot of the PAE vs input power.

It can be seen that the PAE of the PA tends to sharply decrease with decreasing input power. It can also be seen from Figure 3 that maximum PAE is achieved near compression. Therefore this type of power control has the advantage of good linearity and predictive control at the expense of decreased PAE.

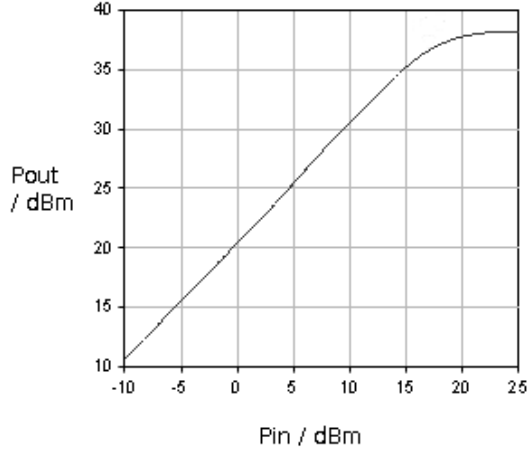


Figure 2. P_{out} vs P_{in} simulation for RF power amplifier.

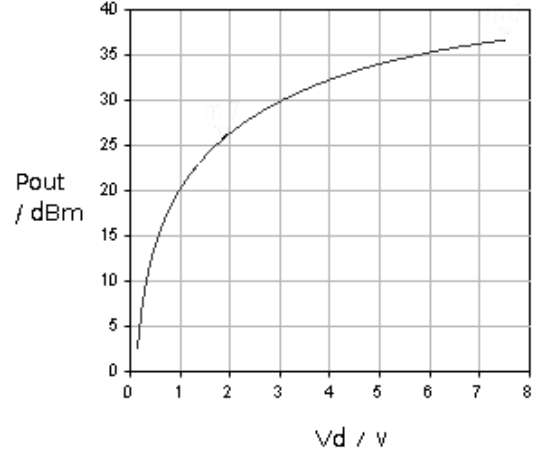


Figure 4. P_{out} vs V_d simulation for RF power amplifier.

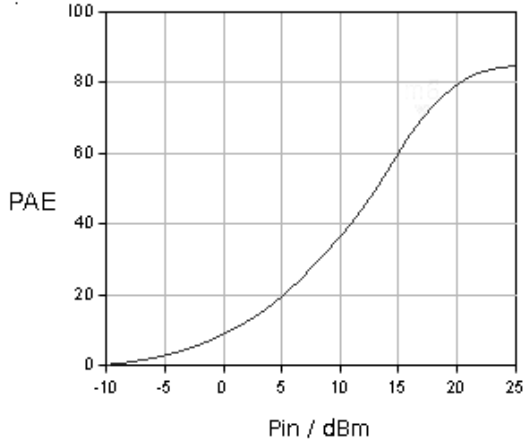


Figure 3. PAE vs P_{in} simulation for RF power amplifier.

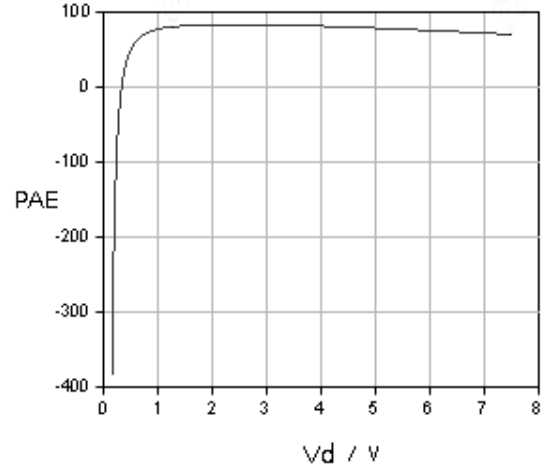


Figure 5. PAE vs V_d simulation for RF power amplifier.

Figure 4 shows the input vs output power curve for the drain bias power control method. Unlike the curve shown in Figure 2 the output vs input power is not linear, but rather logarithmic. Therefore drain bias control requires more complicated design to handle large power delta (Δ) at a lower output power range. In contrast to Figure 3 the PAE vs drain voltage (V_d) does not decrease significantly until the lower output power range (< 20 dBm) is reached. This is one advantage of variable drain bias control: the ability to maintain a steady PAE for the upper power range. The total component cost required for the drain power control strategy is significantly lower than most commercially available 0.25 dB digital step attenuators, requiring only a MOSFET, op-amp and a few passives. However, this is at the cost of reduced linearity of the output power vs the desired set-point as mentioned.

Figure 6 shows the resulting PA module that was built. This is a single-ended class-AB RF power amplifier.

The PA module's input section consists of a 2 element LC matching and stabilization network, while the output section consists of a matching and 5'th order harmonic LC π -filtering network. All DC bias ports are isolated from the RF AC path by a bias-tee and are accessible from the DIL (dual in-line) header at the top of the PA module. These DC bias ports are used to ultimately control the RF output power of the PA by utilizing external digital control circuitry. In addition, four M3 screws are used to mount the PA to a suitable heatsink to ensure the thermal stability of the PA. Figure 7 shows the directional coupling and RMS power detection stage of the prototype which is critical for controlling the RF output power of the system.

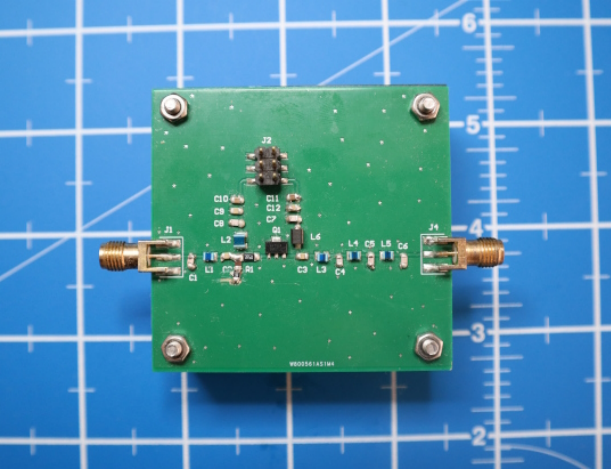


Figure 6. Resulting PA module that was built.

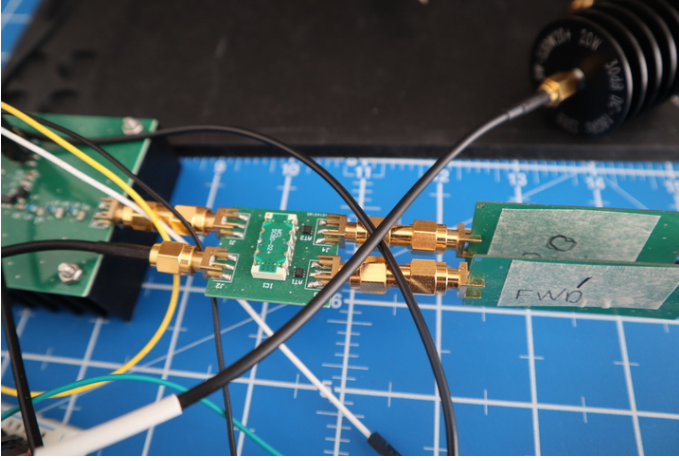


Figure 7. Directional coupler and RMS-log power detector section for the RF forward power.

Both forward and reverse power can be determined using this set-up, but only forward power is required to control the RF output power. The control algorithm used in the feedback loop was a simple, ramp/proportional controller with either the drain voltage or input attenuation being adjusted in fixed steps in order to achieve a specific output power. Using this setup, a power setpoint step size of 0.5 dB was achieved within 20% error. The PA module itself was able to achieve a PAE of 54% in class-AB mode and a large signal gain of 10.89 dB at a maximum RF output power of +30 dB. Figure 8 shows the resultant measured output power vs setpoint in dBm for the drain bias control strategy. A similar graph of the measured output vs setpoint power was obtained for the attenuation base control method as shown in Figure 9.

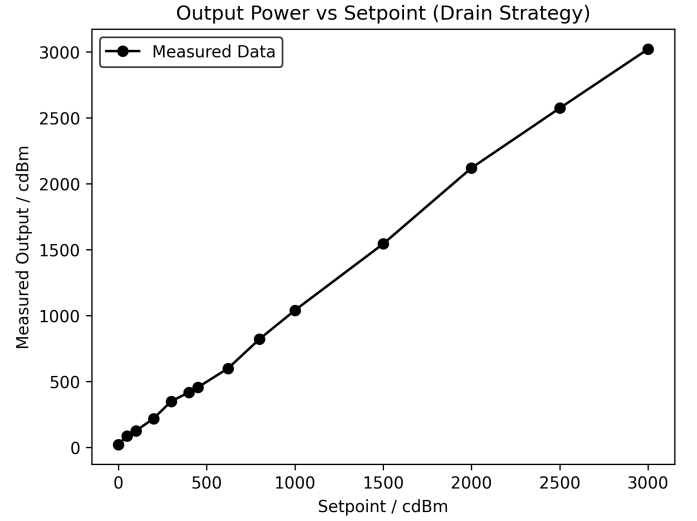


Figure 8. RF output power vs set point for the drain power control strategy.

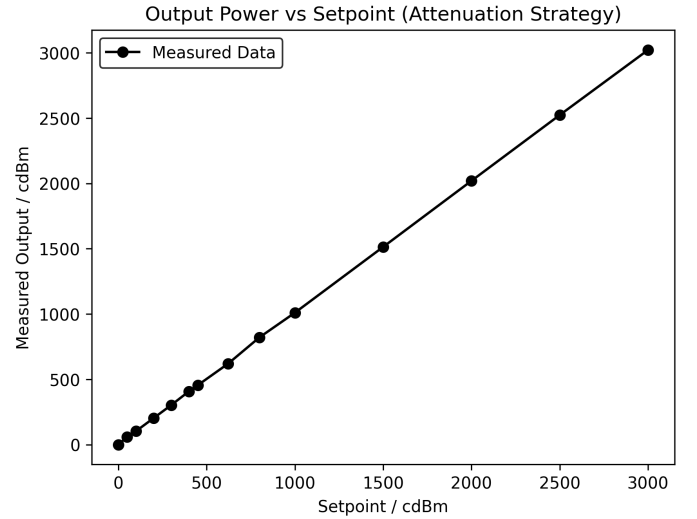


Figure 9. RF output power vs set point for the attenuation based power control strategy.

As shown in Figure 8 the measured output power is roughly proportional to the target set point, but the trend is clearly not perfectly linear. However, this is to be expected since earlier simulations demonstrated that the relationship between P_{out} and drain voltage is not a simple linear function (Figure 4), and that the rate of change of output power, for lower power levels, is much higher and therefore more difficult to control.

In general, an RF power control strategy can either be open-loop or closed-loop and determines the mechanism that achieves RF output power control. Closed-loop power control strategies often make use of RF sensing elements such as a directional coupler and a RMS logarithmic power sensor [7] and are therefore more complex and costly to implement. The main drawback of

the closed-loop approach is the insertion loss due to the RF directional coupling element. Additionally, the loop gain of the feedback network can vary significantly due to the non-linear characteristics of the PA [8], potentially introducing control stability issues. However, closed-loop power control generally has the advantage of improved accuracy compared to open-loop methods.

VI. CONCLUSION

The total component cost required for the drain power control strategy is significantly lower than most commercially available 0.25 dB digital step attenuators, requiring only a MOSFET, op-amp and a few passives. However, this is at the cost of reduced linearity of the output power vs the desired set-point. Therefore, it is up to the end user to decide whether cost, linearity, or settling time are more important, since this will ultimately decide which power control strategy to choose for their particular application. According to the performance data collected for the power amplifier module, all basic power control performance requirements were met for both power control strategies that were tested, with a maximum power step size deviation ΔD of less than 20 %, a maximum amplitude accuracy of 0.9 dB and a maximum settling time (T_S) of 550 ms in the worst case for both power control strategies. In summary, the system was able to output a 169.4 MHz CW tone with an output power ranging from 0 to 30 dBm in 0.5 dB steps, with a PAE of ≈ 54 % and a large signal gain of 10.89 dB. These were the main achievements of the VARPA project.

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